

Juan Esteban Villegas Delgado

Ph.D. Electrical Engineering
Mechatronics Engineer, M.Sc. Microsystems Engineering
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EDUCATION

Escuela de Ingeniería de Antioquia (EIA) University, Medellín, ANT, Colombia

Bachelor of Sciences, Mechatronics Engineering, December 2011

Thesis: “Model of solar powered water pumping systems for small distribution networks on rural arid regions in Colombia”, work carried with the Sustainability Research Group at EIA University

Khalifa University (Masdar Institute), Abu Dhabi, AD, United Arab Emirates
Master of Science, Microsystems Engineering, May 2018

Thesis: “Studies of materials for planar integrated photonics: PDMS and amorphous Silicon”, work carried at the Masdar Institute, UAE, in collaboration with MIT.

New York University, New York City, USA & Abu Dhabi, UAE

Philosophy Doctor, Electrical Engineering, October 2023

Thesis: “Integrated Silicon Photonics for Physical Layer Security Applications”, work carried at the Center for Cyber Security at NYU Abu Dhabi.

PROFILE

Senior Quantum Hardware Engineer specializing in the full-stack architecture, cleanroom process optimization, and cryogenic bring-up of superconducting quantum processing units (QPUs) and integrated photonic systems. In my current role, I lead superconducting circuit design and act as technical fabrication reviewer—bridging electromagnetic/FEM layout with wafer-scale Design-for-Manufacturing (DFM), yield optimization, and device reliability. I drive superconducting thin-film and junction process control across Aluminum, Niobium, and Tantalum platforms, tracking critical temperature (T_c), sheet resistance uniformity, Josephson junction critical current density (J_c) mapping, dielectric loss-tangent reduction, and long-term resonator Q -factor and coherence stability through full multi-qubit QPU bring-up and automated calibration.

Beyond device physics, I bring over a decade of industrial engineering and operational leadership—beginning with leading multidisciplinary engineering teams in commercial high-speed boat and naval ship design, where I managed client requirements, statutory compliance, and cross-team conflict resolution under strict delivery schedules. In cleanroom and hardware laboratory environments, I author and audit Standard Operating Procedures (SOPs), train technicians and junior engineers in nanofabrication and metrology protocols, and continuously integrate emerging fabrication paradigms including 3D micro-airbridges and additive/hybrid manufacturing techniques.

At the organizational level, I manage complex hardware project schedules, oversee cleanroom equipment procurement, negotiate vendor service and maintenance contracts, and coordinate external supplier and foundry deliverables. I actively contribute to team growth and governance by conducting technical interviews and candidate evaluations for entry- and mid-level engineering positions, mentoring staff, and aggregating technical milestones and KPIs for formal performance reviews. Proficient in English, Spanish, and French.

TECHNICAL SKILLS

Quantum Hardware Design, EDA & AI-Driven Automation: Full-stack circuit quantum electrodynamics (cQED) and integrated photonic architecture; Hamiltonian modeling and energy-participation quantization (pyEPR, scqubits, QuTiP); programmatic physical layout generation, PDK development, and automated Design-for-Manufacturing (DFM) verification (KLayout Python API); AI/ML-assisted inverse design, automated parameter sweeps, and closed-loop electromagnetic/optical design-space optimization.

QPU Bring-Up, Cryogenic RF & Automated Calibration: End-to-end cryogenic RF/microwave packaging and multi-qubit QPU characterization below 20 mK; operation, troubleshooting, and preventative maintenance of dilution refrigerators and ultra-high vacuum (UHV) systems; development of automated data-acquisition, pulse-level control, and AI-integrated QPU calibration and benchmarking pipelines (Qibo/Qibocal, Quantify, Qiskit) for single- and two-qubit gate tuning, crosstalk mitigation, and automated tracking of qubit coherence (T_1, T_2^*) and resonator internal quality factors (Q_i).

Cleanroom Microfabrication & Process Control: Wafer-scale process development, fabrication readiness review, and yield optimization for superconducting (Al, Nb, Ta) and silicon photonic (Si, Si₃N₄) platforms; Dolan/Manhattan shadow-evaporated Josephson tunnel junctions, 3D micro-airbridges, and heterogeneous 2D material integration; thin-film deposition (e-beam and thermal evaporation, magnetron sputtering, ALD, PECVD, LPCVD, electroplating), dry/wet etching (ICP-RIE, DRIE, FIB, wet chemical etching), and lithography (Electron-Beam, laser direct-write, and optical lithography); cleanroom SOP authoring and additive manufacturing.

Photonic, Superconducting & Material Metrology: Room-temperature and cryogenic wafer probing, four-point sheet-resistance and critical current density (J_c) mapping, critical temperature (T_c) and dielectric loss-tangent characterization; high-speed opto-electronic testing, Optical Vector/Network Analysis, insertion-loss and polarization metrology, and DC I - V curve tracing; Scanning Electron Microscopy (SEM-EDS), Atomic Force Microscopy (AFM), X-Ray Diffraction (XRD), Raman spectroscopy, spectroscopic ellipsometry, optical interferometry/profilometry, ICP-MS, and Ion Chromatography.

Programming & Software Engineering: Python (scientific computing, hardware control, ML/automation), C++, C#, MATLAB, LabVIEW, Git, Docker, Linux.

Simulation & Quantum Frameworks: Ansys (HFSS, Q3D), COMSOL Multiphysics, Lumerical (FDTD, MODE), MEEP, QuTiP, scqubits, pyEPR, Qiskit, Qibo (qibolab/qibocal), Quantify.

CAD & Physical Layout: KLayout (scripted EDA/PCell automation), SolidWorks, Autodesk Inventor, AutoCAD, Rhinoceros.

CURRENT PROJECTS

New Materials for Superconductive Quantum Electrodynamics: *2023-Current* - Process development, thin-film growth qualification, and cryogenic RF characterization of high-impedance (high kinetic inductance L_k) superconducting materials for next-generation QPUs, targeting sheet-resistance uniformity, critical temperature (T_c) control, and low dielectric loss.

Principal Investigator: Dr. Frederico Brito, TII, UAE.

Design and Test of Superconductive Quantum Processing Units: *2023-Current*

- Full-stack circuit quantum electrodynamics (cQED) architecture, FEM simulation, and fabrication review of multi-qubit QPUs—including high- Q coplanar waveguide resonators, tunable and fixed couplers, 3D airbridges, and Al/AIO_x/Al Josephson tunnel junctions (J_c targeting and yield optimization).

Principal Investigator: Dr. Frederico Brito, TII, UAE.

SELECTED COMPLETED PROJECTS

8-Qubit Fixed-Frequency All-Microwave Superconducting QPU: 2025-2026

- Led the end-to-end Hamiltonian parameter calculation, electromagnetic simulation, and physical layout of an 8-qubit fixed-frequency all-microwave quantum processor. Coordinated cleanroom manufacturing, DFM reviews, and wafer qualification with the fabrication team; led cryogenic bring-up with the measurement lab team and executed calibration experiments to produce high-fidelity single- and two-qubit microwave gates.

Direct Manager: Dr. Frederico Brito, TII, UAE.

Low-Loss Superconducting Platform Based on Tantalum (Ta) Encapsulation: 2025-2026

- Designed experimental test vehicles and microwave resonator metrology structures to evaluate Tantalum-encapsulated superconducting thin films aimed at suppressing surface oxide formation and two-level system (TLS) dielectric losses. Led experimental design, oversaw thin-film process characterization (T_c , sheet resistance, and internal quality factor Q_i stability), and reviewed scientific data analysis.

Direct Manager: Dr. Frederico Brito, TII, UAE.

Quantum Computing Cryogenic & Cleanroom Infrastructure: 2023-2024

- Commissioned, qualified, and established operational SOPs for the dilution refrigerator measurement facility and cleanroom test infrastructure supporting the first domestically developed quantum processors in the UAE.

Principal Investigators: Prof. Andrew Lutken and Dr. Frederico Brito, TII, UAE.

High-Speed Mode and Polarization Encryption for Optical Communication Networks: 2021-2023

- Designed and fabricated integrated silicon photonic devices and high-speed optical testbeds for mode- and polarization-division multiplexed encryption and Optical Physical Unclonable Function (OPUF) key distribution.

Principal Investigator: Dr. Mahmoud Rasras, NYUAD, UAE.

Heterogeneously Integrated Layered Materials for Silicon Photonics: 2021-2024

- Designed and fabricated planar silicon photonic circuits engineered for back-end-of-line (BEOL) heterogeneous integration with 2D layered materials for high-speed photodetectors and modulators.

Principal Investigator: Dr. Mahmoud Rasras, NYUAD, UAE.

Plasmonic-Enhanced Physically Unclonable Functions: 2020-2021 - Developed nanofabrication processes to integrate plasmonic resonant structures onto silicon photonic waveguides, generating tamper-evident stochastic scattering responses for hardware cryptography.

Principal Investigator: Prof. Mahmoud Rasras, NYUAD, UAE.

PM2.5 Airborne Pollutant Characterization Over the UAE: 2018-2019 - Led multi-site sampling, SEM-EDS/ICP-MS metrology, and source apportionment of PM2.5 particulates across the UAE. Authored policy recommendations for government environmental agencies and trained technical personnel on standardized measurement SOPs.

Direct Manager: Prof. Marouane Temimi, Khalifa University, UAE.

On-Board Load Computers Technical Specification: 2015-2016 - Authored the

statutory classification rules and certification test standards for shipboard load/stability computers and marine electronic systems across European inland waterways.

Direct Manager: Mr. Jean Michel Chatelier, Bureau Veritas, Belgium.

Naval Datalink System: 2014-2015 - Led system architecture, supplier specification, and hardware/firmware prototyping for a tactical naval datalink platform (OSI layers 2–6) deployed by the Colombian Navy.

Direct Manager: (R) Rear-Admiral LH (OF-6) Gustavo Perez, COTECMAR - ARC, Colombia.

Riverine Patrol Craft (LPR-MkII): 2012-2014 - Led multidisciplinary engineering design covering updated hull hydrodynamics, structural layout, electrical/electronic system integration, and shipyard production support for a rapid riverine patrol craft series.

Direct Manager: Commander (OF-4) Ricardo Mejía, COTECMAR - ARC, Colombia.

WORK EXPERIENCE

Postdoctoral Researcher —

Technology Innovation Institute

Superconducting Circuit Lead

Designer & Fabrication Reviewer

September 2023 - to date

Abu Dhabi, UAE

Lead the full-stack microwave circuit design, fabrication review, and cryogenic bring-up of multi-qubit superconducting Quantum Processing Units (QPUs), alongside operational management of an ISO Class-5 cleanroom and dilution refrigerator facility (1.65 mW combined cooling power at 20 mK).

- **Circuit Design & Fabrication Readiness Review:** Lead the electromagnetic/FEM design and automated physical layout of multi-qubit superconducting processors; serve as primary technical fabrication reviewer enforcing Design-for-Manufacturing (DFM) rules, wafer-level yield optimization, and process window control. Design and develop EDA integration tools for superconductive circuits. Develop and coordinate trainings for QPU design, fabrication and calibration.
- **Superconducting Process Control & Reliability Metrics:** Drive thin-film (Al, Nb, Ta) and Josephson junction process qualification, tracking film conductivity and critical temperature (T_c), sheet resistance uniformity, critical current density (J_c) wafer mapping, dielectric loss-tangent reduction, and long-term resonator Q -factor and qubit coherence stability.
- **QPU Bring-Up & Automated Calibration:** Lead cryogenic RF packaging, cooldown characterization, and full-stack QPU bring-up, developing automated single- and two-qubit calibration and benchmarking pipelines.
- **Cleanroom Operations, SOPs & Training:** Author, review, and enforce cleanroom Standard Operating Procedures (SOPs) for lithography, etching, thin-film deposition, and 3D airbridge integration; train and supervise technicians and junior engineers on tool operation, safety, and metrology protocols.
- **Procurement, Suppliers & Service Contracts:** Manage cleanroom and cryogenic capital/consumable procurement, qualify external foundries and specialty material suppliers, and oversee vendor preventative maintenance and service contracts for critical fabrication and cryogenic infrastructure.
- **Project Governance, Hiring & Performance Reviews:** Manage multi-project hardware delivery schedules, conduct technical screenings and interviews for entry- and mid-level engineering candidates, mentor staff, and aggregate project KPIs and technical milestones for formal performance reviews.

Graduate Research Intern

July 2022 - October 2022

Technology Innovation Institute

Abu Dhabi, UAE

Designed, taped out, and characterized integrated Silicon Nitride (Si_3N_4) photonic chips for quantum single-photon sources and commercial-grade Quantum Random Number Generators (QRNGs), coordinating foundry DFM compliance and optical test-bench automation.

Technology and Data Leader

January 2018 - June 2020

Airborne

Abu Dhabi, UAE

Directed the product engineering and hardware/software integration division developing commercial indoor navigation systems for visually impaired users. Managed product roadmaps, cross-functional engineering schedules, client and stakeholder requirements, and end-to-end prototype-to-product validation.

Research Engineer

August 2018 - April 2019

Khalifa University

Abu Dhabi, UAE

Led analytical metrology and high-resolution physical/chemical characterization workflows using SEM-EDS, Ion Chromatography (IC), and ICP-MS. Established laboratory quality-control SOPs, calibrated analytical instrumentation, and authored technical compliance reports.

Development Engineer

June 2015 - July 2016

Bureau Veritas

Antwerp, BE

Authored statutory engineering rules, regulatory SOPs, and technical standards for commercial inland navigation vessels, specializing in shipboard electrical distribution, automation, and electronic control systems. Interfaced directly with shipyards, equipment manufacturers, and regulatory bodies, and developed and delivered technical certification courses for industry professionals.

Marine Designer and Researcher —

COTECMAR

Engineering Team Lead

February 2012 - June 2015

Cartagena, CO

Led multidisciplinary engineering teams across the Naval Architecture and Electrical/Electronics divisions for the industrial design, manufacturing, and commissioning of high-speed commercial and patrol vessels.

- **Team & Schedule Leadership:** Managed engineering design teams, master project schedules, and cross-departmental conflict resolution between design, shipyard production, and quality assurance units.
- **Supplier, Procurement & Client Management:** Defined technical specifications for shipboard electronics and power systems, evaluated vendor bids, managed supplier deliverables, and led client design reviews and sea-trial acceptance testing.
- **R&D & Prototyping:** Led applied R&D in ruggedized naval electronics and telecommunications, taking tactical marine datalink protocols from architecture through hardware prototyping and field deployment.

**TEACHING
EXPERIENCE**

I have previous experience teaching in different environments, including training new volunteers at a poverty eradication NGO in Colombia and Belgium, training adult professionals in engineering standards and practices in Belgium, and conducting voluntary Spanish workshops at NYUAD. I have acted as a Teaching Assistant in the following university-level courses:

- *ENGR-UH-1000 Computer Programming for Engineers [Undergraduate]* (Fall-21), Mohammed Eid, New York University, Abu Dhabi

- *MIC630 Fundamentals to Photonics [Graduate]* (Fall-17), Marcus Dahlem, Khalifa University, Abu Dhabi
- *FDN475 Photonic Technology [Undergraduate]* (Spring-18), Marcus Dahlem and Mahmoud Rasras, Khalifa University, Abu Dhabi

CURRENT INTERESTS

System automation and robotics, Integrated Optics, Silicon Photonics, Hardware Security, Cryptography, Additive Manufacturing, Quantum Computing

NON WORK EXPERIENCE

Founding member of EERF (Research Forum for early career scientists at NYUAD), and its Chair in the Spring 2023 semester. Former Member of the research group in Alternative Materials and Processes at the University EIA (Colombia) until 2011, the Nano optics and Optoelectronics research laboratory in Masdar (UAE) until 2020 and the Photonics Research Lab at NYUAD (Abu Dhabi) until 2024. Volunteer (July 2013 - September 2016) with TECHO (NGO based in Chile working on extreme poverty eradication), manager of the Construction and Logistics team in Cartagena, Colombia (December 2014 to July 2015), and National Coordinator of Volunteering in Brussels, Belgium (September 2015 to July 2016). Volleyball, climbing and sailing enthusiast. Rugby player. Member of the Graduate Student Council at NYU Abu Dhabi (from 2019 to 2022). Member of the IEEE Photonics Society, Optica and the Colombian Optical Society.

CONTINUOUS FORMATION

Superconductive Qubit Design, CMC Microsystems, Vancouver, Canada (remote) (December 2024). Silicon Photonics Design, CMC Microsystems, Vancouver, Canada (August 2022). Silicon Photonics Design, Fabrication and Data Analysis, UBCx, edX Online Course (November 2019). Teaching Assistant Training, NYUAD, Abu Dhabi, UAE (December 2021). Scientist teaching Sciences, New York Academy of Sciences (August 2020). Train the trainer (Management and development of adult trainings), Opatija, Croatia (March 2016). Inland Navigation Rules and Statutory Regulations, Antwerp, Belgium (November 2015). Solar Energy, DelftX, edX Online Course (December 2015). Electromagnetic Compatibility, Cartagena, Colombia (June 2014).

ACADEMIC REFERENCES

Prof. Dr. Mahmoud Rasras, Associate Professor, New York University, Abu Dhabi, UAE, mrasras@nyu.edu, +971 (0) 567923399
Prof. Dr. Jaime Viegas, Associate Professor, Khalifa University, Abu Dhabi, UAE. jaime.viegas@ku.ac.ae, +971 (0) 566986928
Prof. Dr. Frederico Brito, Director Quantum Hardware, Quantum Research Center, TII, Abu Dhabi, UAE. frederico.brito@tii.ae

PERSONAL REFERENCES

Dr. Catherine Ramirez, Research Engineer at Syroco, Marseille, France, lcramirev@gmail.com, +33 665003530
Dr. Dan Cracan, Analog Mixed Signal Verification Engineer, Apple, UK, dan.cracan@gmail.com, +44 7930 164 762
Dr. Florent Ravaux, Senior Postdoctoral Researcher, TII, Abu Dhabi, UAE, florent.ravaux@tii.ae, +971 (0) 509009608

PUBLICATIONS

IN JOURNALS

- Alkhazaleh, A., Villegas, J., Ravaux, F., & Zharinov, A. (2026). *Tantalum-encapsulated niobium superconducting resonators: High internal quality factor and improved temporal stability via surface passivation*. Retrieved from <https://arxiv.org/abs/2604.09050>
- Villegas, J. E., Tamalampudi, S. R., & Rasras, M. (2025, Apr). High-speed polarization control in integrated silicon photonics. *Opt. Express*, 33(8), 18477–18491. Retrieved from <https://opg.optica.org/oe/abstract.cfm?URI=oe-33-8-18477> doi: 10.1364/OE.523366
- Villegas, J. E., & Rasras, M. (2024, Mar). A compact mode controller (switch) using an inverse-designed 2-mode y-junction. *J. Lightwave Technol.*, 42(6), 2083–2089. Retrieved from <https://opg.optica.org/jlt/abstract.cfm?URI=jlt-42-6-2083>
- Tamalampudi, S. R., Villegas, J. E., Dushaq, G., Sankar, R., Paredes, B., & Rasras, M. (2023b). *A high-speed waveguide integrated inse photodetector on sin photonics for nir applications*.
- Villegas, J. E., Jimoh, Y. O., & Rasras, M. (2023). Polarization encryption system using commercial leds for additive manufacturing. *Opt. Continuum*, 2(4), 783–792. doi: 10.1364/OPTCON.480991
- Villegas, J. E., Paredes, B., & Rasras, M. (2021a). Experimental studies of plasmonics-enhanced optical physically unclonable functions. *Opt. Express*, 29(20), 32020–32030. doi: 10.1364/OE.437636
- Tamalampudi, S. R., Villegas, J. E., Dushaq, G., Sankar, R., Paredes, B., & Rasras, M. (2023a). High-speed waveguide-integrated inse photodetector on sin photonics for near-infrared applications. *Advanced Photonics Research*, 4(11), 2300162. doi: 10.1002/adpr.202300162
- Zafar, H., Paredes, B., Villegas, J., Rasras, M., & Pereira, M. F. (2023, Jun). O-band te- and tm-mode densely packed adiabatically bent waveguide arrays on the silicon-on-insulator platform. *Opt. Express*, 31(13), 21389–21398. doi: 10.1364/OE.493077
- Zafar, H., Paredes, B., Taha, I., Villegas, J. E., Rasras, M., & Pereira, M. F. (2023). Compact and broadband adiabatically bent superlattice-waveguides with negligible insertion loss and ultra-low crosstalk. *IEEE Journal of Selected Topics in Quantum Electronics*, 29(6: Photonic Signal Processing), 1-9. doi: 10.1109/JSTQE.2023.3241617
- Tamalampudi, S. R., Dushaq, G., Villegas, J. E., Paredes, B., & Rasras, M. S. (2023). A multi-layered gagete electro-optic device integrated in silicon photonics. *Journal of Lightwave Technology*, 1-7. doi: 10.1109/JLT.2023.3237818
- Dushaq, G., Villegas, J. E., Paredes, B., Tamalampudi, S. R., & Rasras, M. S. (2023a). Anisotropic van der waals 2d geas integrated on silicon four-waveguide crossing. *J. of Lightwave Technology*, 41(6), 1784-1789. doi: 10.1109/JLT.2022.3229069

Tamalampudi, S. R., Dushaq, G., Villegas, J. E., Paredes, B., & Rasras, M. S. (2022). Si₂Te₃ photodetectors for optoelectronic integration at telecommunication wavelengths. *IEEE Journal of Selected Topics in Quantum Electronics*, 28(3), 1-7. doi: 10.1109/JSTQE.2021.3126612

Zafar, H., Zhai, Y., Villegas, J. E., Ravaux, F., Kennedy, K. L., Pereira, M. F., ... Anjum, D. H. (2022, Mar). Compact broadband (o, e, s, c, l & u bands) silicon te-pass polarizer based on ridge waveguide adiabatic s-bends. *Opt. Express*, 30(6), 10087–10095. doi: 10.1364/OE.452823

Dushaq, G., Paredes, B., Villegas, J. E., Tamalampudi, S. R., & Rasras, M. (2022b, May). On-chip integration of 2d van der waals germanium phosphide (gep) for active silicon photonics devices. *Opt. Express*, 30(10), 15986–15997. doi: 10.1364/OE.457242

Paredes, B., Mohammed, Z., Villegas, J., & Rasras, M. (2021a). Dual-band (o and c-bands) two-mode multiplexer on the soi platform. *IEEE Photonics Journal*, 13(3), 1-9. doi: 10.1109/JPHOT.2021.3075292

Tamalampudi, S. R., Dushaq, G., Villegas, J. E., Rajput, N. S., Paredes, B., Elamurugu, E., & Rasras, M. S. (2021). Short-wavelength infrared (swir) photodetector based on multi-layer 2d gage. *Opt. Express*, 29(24), 39395–39405. doi: 10.1364/OE.442845

Alfakes, B., Garlisi, C., Villegas, J., Al-Hagri, A., Tamalampudi, S., Rajput, N. S., ... Chiesa, M. (2020). Enhanced photoelectrochemical performance of atomic layer deposited hf-doped zno. *Surface and Coatings Technology*, 385, 125352. doi: 10.1016/j.surfcoat.2020.125352

Odeh, M., Twayana, K., Sloyan, K., Villegas, J. E., Chandran, S., & Dahlem, M. S. (2019). Mode sensitivity analysis of subwavelength grating slot waveguides. *IEEE Photonics Journal*, 11(5), 1-10. doi: 10.1109/JPHOT.2019.2939088

Alfakes, B., Villegas, J., Apostoleris, H., Devarapalli, R., Tamalampudi, S., Lu, J.-Y., ... Chiesa, M. (2019). Optoelectronic tunability of hf-doped zno for photovoltaic applications. *The Journal of Physical Chemistry C*, 123(24), 15258-15266. doi: 10.1021/acs.jpcc.9b02253

CONFERENCE PROCEEDINGS

Ravaux, F., Villegas, J. D., Alkhazaleh, A., Kazieva, T., & Zharinov, A. (2026). Air-bridges mechanical stability improvement using stiffening techniques for superconducting quantum circuit application. In *2026 ieee international conference on quantum computing and engineering*.

Villegas, J. E., Tamalampudi, S. R., & Rasras, M. S. (2024). A high-speed compact polarization controller in silicon photonics. In *2024 optical fiber communications conference and exhibition (ofc)* (p. 1-3).

Dushaq, G., Villegas, J. E., Paredes, B., Tamalampudi, S. R., & Rasras, M. (2023b). Four-waveguide crossing functions utilizing anisotropic van der waals 2d geas. In *Cleo 2023* (p. 1-2). Optica.

- Tamalampudi, S. R., Villegas, J. E., Paredes, B., Dushaq, G., & Rasras, M. S. (2023). Integrated sin/inse waveguide photodetector for o-band applications. In *Cleo: Science and innovations* (p. JW2A. 41). Optica.
- Villegas, J. E., & Rasras, M. (2023). Multimode silicon y-splitter with in-phase and off-phase te0 outputs. In *2023 ieee silicon photonics conference (siphotonics)* (p. 1-2). doi: 10.1109/SiPhotonics55903.2023.10141955
- Tamalampudi, S., Dushaq, G., Villegas, J., & Rasras, M. (2023). Integrated 2d gage electro-optic phase shifter in silicon photonics. In *International conference on meta-materials, photonic crystals and plasmonics* (pp. 227–229).
- Paredes, B., Mohammed, Z., Villegas, J. E., & Rasras, M. (2021c). Ultra-compact ultra-broadband two-mode transverse-electric based swg multiplexer demonstrated at 64 gbps. In *Ecoc* (p. 1-3). Optica.
- Villegas, J. E., Dushaq, G., & Rasras, M. (2022). Inverse design of a 4-way crossing for integration of anisotropic materials on silicon waveguides. In *Frontiers in optics + laser science 2022 (fio, ls)* (p. JTu5A.58). Optica Publishing Group.
- Dushaq, G., Paredes, B., Villegas, J. E., Tamalampudi, S. R., & Rasras, M. (2022a). Hybrid integration of multilayers gep on a silicon photonics platform for optoelectronic application. In *Conference on lasers and electro-optics* (p. JW3A.33). Optica Publishing Group.
- Paredes, B., Mohammed, Z., Villegas, J., & Rasras, M. (2021b). An ultra-broadband two-mode transverse-electric multiplexer in soi platform. In *2021 optical fiber communications conference and exhibition (ofc)* (p. 1-3).
- Villegas, J. E., Paredes, B., & Rasras, M. (2021b). Performance of silicon opufs under variable input losses. In *2021 conference on lasers and electro-optics europe & european quantum electronics conference (cleo/europe-eqec)* (p. 1-1). doi: 10.1109/CLEO/Europe-EQEC52157.2021.9542040
- Mohammed, Z., Paredes, B., Villegas, J., & Rasras, M. (2021). An ultra-compact cmos compatible mmi based 1310/1550 nm wavelength (de) multiplexer. In *2021 european conference on optical communication (ecoc)* (p. 1-3). doi: 10.1109/ECOC52684.2021.9606089
- Odeh, M., Twayana, K., Sloyan, K., Villegas, J. E., Chandran, S., & Dahlem, M. S. (2018). Slotted subwavelength grating waveguides for compact optofluidic sensors. In *2018 international conference on optical mems and nanophotonics (omn)* (p. 1-2). doi: 10.1109/OMN.2018.8454591
- Villegas, J. E., Odeh, M., & Dahlem, M. S. (2017). Integrated gradient-index planar optofluidic polymer waveguide. In *Frontiers in optics 2017* (p. JTu2A.18). Optica Publishing Group.